

AZURE BATCH FOR AN ENGINEERING SIMULATION WORKLOAD

Azure Services List and System Flow Chart

24CC3046-P075 | High Performance Computing | T075, T177 | 8 Students

1. Project Overview

This project demonstrates the use of Microsoft Azure Batch for a High Performance Computing (HPC) engineering simulation workload. A target workload of 300 simulation jobs is divided into individual Batch tasks and distributed across a pool of cloud compute nodes. Results are stored centrally in Azure Storage and the workload is monitored through Azure services.

2. Azure Services Used

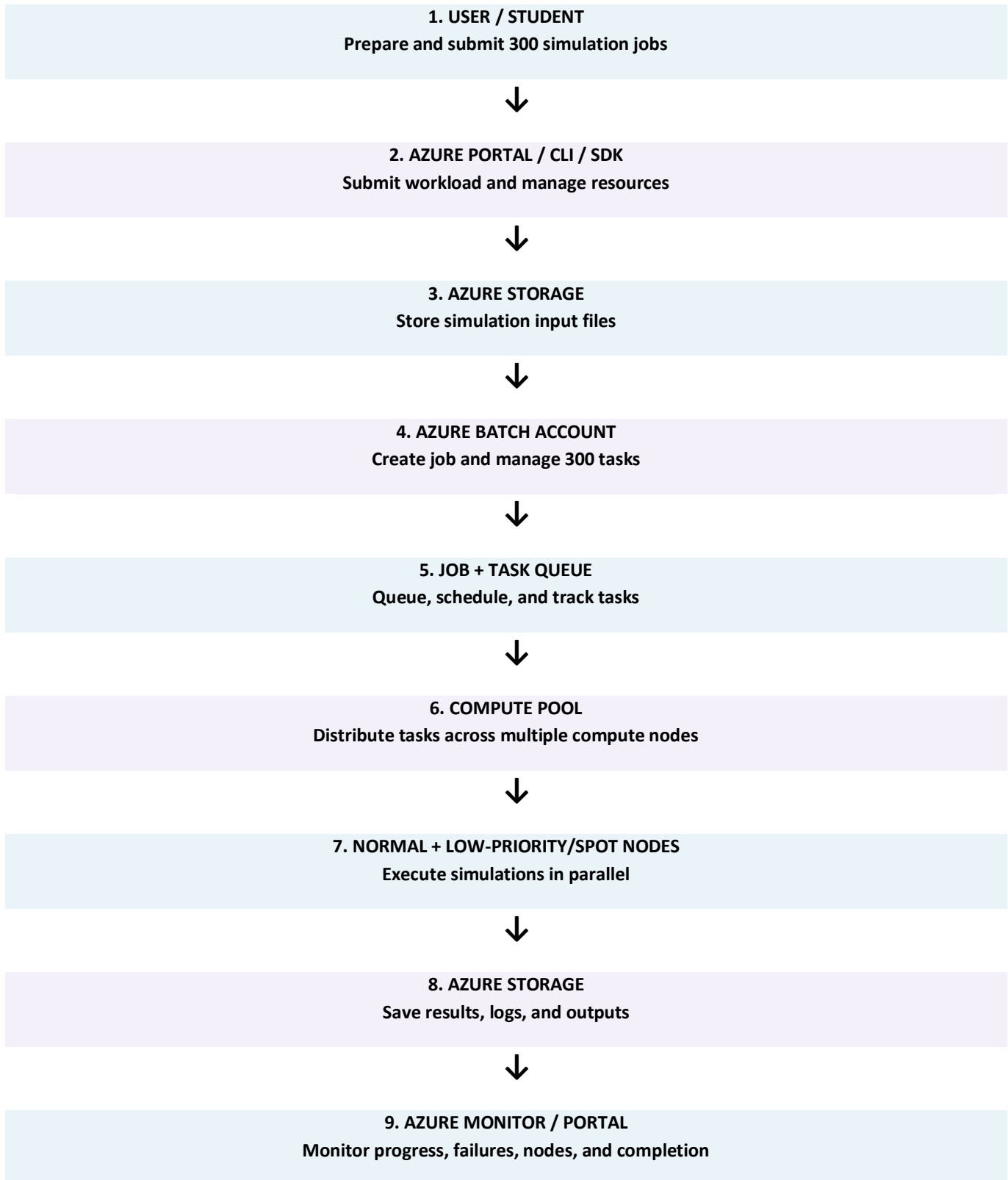
No.	Azure Service	Purpose in the Project
1	Azure Batch	Creates and manages compute pools, jobs, tasks, scheduling, parallel execution, and task status.
2	Azure Storage Account	Stores simulation input files, output results, logs, and other persistent data.
3	Azure Batch Compute Pool / Virtual Machines	Provides the CPU and memory resources on which simulation tasks execute.
4	Azure Monitor	Provides monitoring, diagnostics, performance information, and failure visibility.
5	Azure Resource Group	Organizes the Batch, Storage, networking, and monitoring resources for the project.
6	Azure Virtual Network (VNet)	Provides controlled network connectivity and isolation for cloud resources when required.
7	Microsoft Entra ID	Provides identity and access management for users and Azure resources.
8	Azure Key Vault	Protects secrets, keys, and credentials when the implementation requires them.
9	Azure Portal	Used to create, configure, monitor, and demonstrate the Azure resources.
10	Azure CLI / Azure SDK	Supports automation of Batch jobs, task submission, storage operations, and resource management.

3. Core Services for the Student Implementation

- Azure Batch – main workload orchestration service.
- Azure Batch Compute Pool – executes the simulation tasks.
- Azure Storage – persistent input, output, and log storage.
- Azure Monitor – monitors workload and resource status.
- Azure Portal – resource configuration and project demonstration.

4. System Flow Chart

The following flow shows how the 300 engineering simulation jobs move through the Azure Batch architecture.



5. Architecture Flow

User → Azure Portal/CLI → Azure Batch → Job & Task Queue → Compute Pool → Parallel Simulation Tasks → Azure Storage → Azure Monitor

6. Detailed Execution Process

Step 1 – Prepare Inputs: Prepare the input parameters and files required by each engineering simulation.

Step 2 – Upload Inputs: Upload simulation inputs to Azure Storage so compute nodes can access the required data.

Step 3 – Create Batch Job: Create an Azure Batch job and define the target pool and task configuration.

Step 4 – Create 300 Tasks: Create one Batch task for each simulation workload and place the tasks in the scheduling queue.

Step 5 – Allocate Compute Nodes: Azure Batch uses the configured compute pool and assigns tasks to available nodes.

Step 6 – Execute in Parallel: Multiple nodes process multiple simulation tasks concurrently.

Step 7 – Handle Interruptions: If a low-priority/Spot node is evicted, the interrupted task can be retried or requeued according to the configured policy.

Step 8 – Store Results: Completed simulation outputs and logs are written to persistent Azure Storage.

Step 9 – Monitor: Azure Monitor and Azure Portal provide visibility into task progress, node status, failures, and completion.

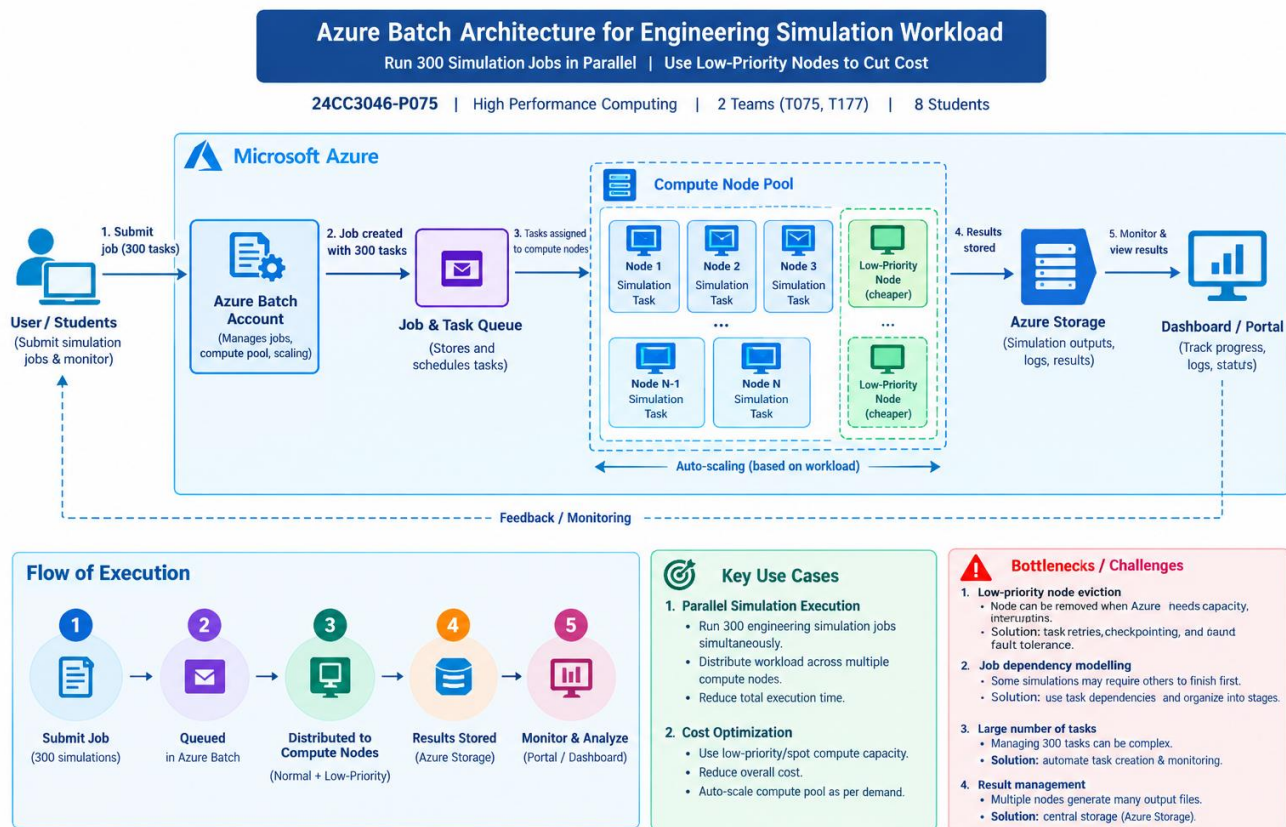
7. Main Challenges and Handling

Challenge	Handling Approach
Low-priority/Spot node eviction	A node may become unavailable while a task is running. Use retry/requeue policies and, where feasible, checkpoint or persist intermediate work.
Job dependency modelling	Dependent tasks must wait for prerequisite tasks. Represent the workflow explicitly, for example: preprocessing → simulation → postprocessing.
Large number of tasks	Managing 300 tasks manually is difficult. Automate task creation, submission, monitoring, and result collection.
Result management	Many nodes can generate many output files. Use centralized Azure Storage with organized input/output/log containers or paths.

8. Expected Outcome

- 300 engineering simulation tasks can be submitted as a single cloud workload.
- Tasks can be distributed across multiple compute nodes and executed concurrently.
- Simulation results and logs are retained in centralized storage.
- Task failures and low-priority/Spot node interruptions can be handled through retries and rescheduling.
- The project demonstrates how cloud-based HPC can provide scalable compute resources and cost-aware execution.

9. Visual Architecture Diagram



9. Note on Azure Cost – Free or Paid

Free or Paid? The project can be demonstrated using a Microsoft Azure free account if the account is eligible and the workload stays within the available free credit and service limits. Azure Batch itself does not have a separate Batch service charge, but the underlying compute nodes, storage, networking, and other resources can incur charges.

New Azure customers may receive USD \$200 credit for the first 30 days, along with specified free monthly amounts for selected services. A short student demonstration may therefore be possible with little or no personal cost when the usage remains within the available credit/limits.

The complete 300-job workload is not guaranteed to be permanently free. The actual cost depends on the virtual-machine size, number of compute nodes, execution time, region, storage, data transfer, and whether Spot/low-priority capacity is used. For a long-running or full-scale HPC workload, paid Azure resources may be required.

Cost Options for This Project

Project Usage	Cost Possibility	Note
Learning / resource creation	Can be free	Use eligible Azure free account/credit and stay within limits.
Small demonstration	May be within free credit	Depends on VM size, node count and runtime.
300-job short demonstration	May fit or may require paid usage	Depends strongly on simulation duration and compute resources.

Continuous / full-scale HPC	Paid	Compute, storage and other Azure resources can generate charges.
Azure Batch service	No separate Batch service fee	Underlying Azure resources are billed separately.

Ways to Control Project Cost

- Use a small VM size for the student demonstration when it is sufficient for the simulation.
- Use Spot/low-priority compute capacity for interruption-tolerant tasks where appropriate.
- Use auto-scaling or controlled pool sizing so unused compute nodes are not kept running.
- Delete or deallocate the compute pool/resources after the demonstration.
- Store only necessary input files, outputs, and logs in Azure Storage.
- Check the Azure Pricing Calculator before running a large or long-duration workload.

Recommended Report Note

Estimated cost: ₹0 during an eligible free-credit demonstration; paid usage depends on Azure resource consumption. Azure Batch itself does not have a separate service fee, while compute, storage, networking and other underlying resources may incur charges.

References for Cost Information

- Microsoft Azure – Free Account: <https://azure.microsoft.com/free/>
- Microsoft Learn – Azure Batch cost management: <https://learn.microsoft.com/azure/batch/batch-azure-services-overview>
- Microsoft Azure – Azure Batch Pricing: <https://azure.microsoft.com/pricing/details/batch/>